Module 01: Systems in Middle School

Learning Objectives

1. Define **Systems** within the context of Middle School.
2. Analyze how Systems interacts with other components of the Active Inference framework.
3. Apply specific constraints of Middle School to the formal definition of Systems.

Introduction

This module explores **Systems**. In the **Middle School** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Systems is a critical component of the 8-part Active Inference spine, bridging the gap between Planning and Agents.

Key Concepts

1. Systems as a Markov Blanket Boundary

How does Systems define the boundary between the agent and the environment?

2. Generative Models of Systems

What parameters involved in Systems must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Systems drive the perception-action loop?

Applications

In Middle School, we see Systems manifest in: * **Specific Example 1**: A video game like Minecraft is a system of interacting parts -- the world generator creates terrain, the physics engine handles gravity and water flow, the crafting system converts raw materials into tools, and the mob AI controls creature behavior. Change one part (like making iron ore rarer) and it ripples through the whole system, affecting what tools you can build, what fights you can win, and how you explore. * **Specific Example 2**: When you write a program with multiple functions that call each other, you have created a system. If your main loop calls a drawPlayer() function that calls a checkCollision() function that calls a updateScore() function, the whole chain depends on each part working correctly. One bug in checkCollision breaks everything downstream -- that is systems thinking applied to code.

Conclusion

Understanding Systems allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.